

SYSTEMS PROJECT MANAGEMENT

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Information Systems

- An **Information System** is a **system** that collects, processes, stores, and disseminates information.
- Information systems are used to support and manage business operations.
- Information systems are used to automate and optimize business processes.
- Examples of information systems include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.

→ first Project



Information Systems

- An **Information System** is a **system** that *collects, processes, stores, and disseminates* information.
- **Information systems** are used to **support** and **manage** business operations.
- **Information systems** are used to **automate** and **optimize** business processes.
- Examples of information systems include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.

processes

computer
scientist



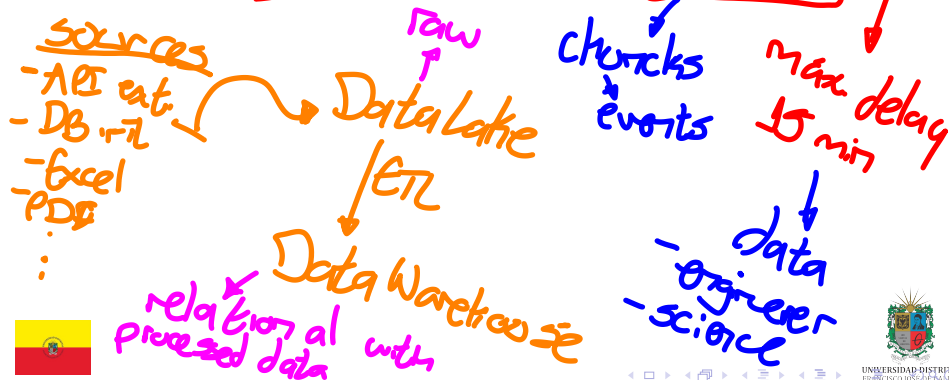
Information Systems

- An **Information System** is a **system** that *collects, processes, stores, and disseminates* information.
- **Information systems** are used to **support** and **manage** business operations.
- **Information systems** are used to **automate** and **optimize** business processes.
- Examples of **information systems** include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.



Data Systems

- A **Data System** is a **system** that **collects**, **processes**, **stores**, and **retrieves data**.
- **Data systems** are used to **store** and **analyze** data.
- Examples of **data systems** include **databases**, **data warehouses**, **data lakes**, **data marts**, **data cubes** and **data streams**.



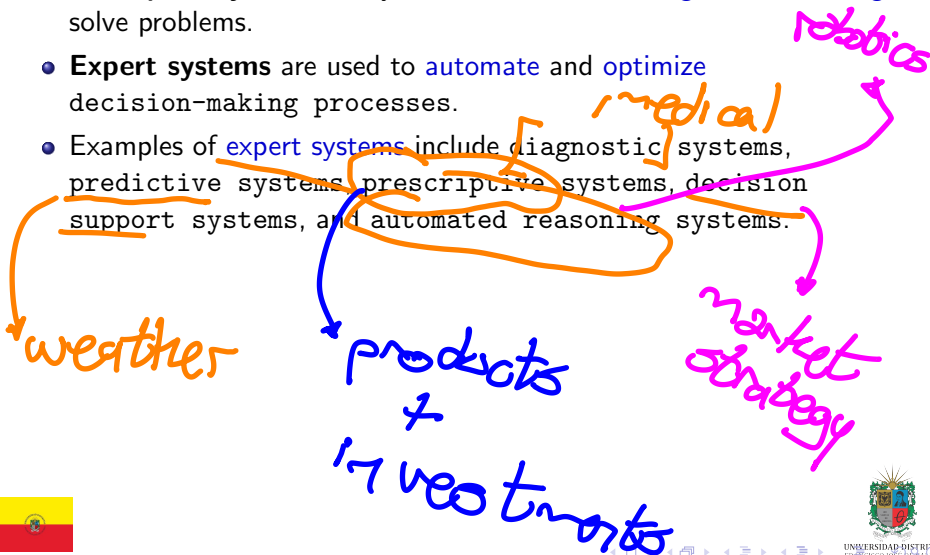
Expert Systems

- An **Expert System** is a system that uses knowledge and reasoning to solve problems. *90's*
- **Expert systems** are used to automate and optimize decision-making processes. *Partial*
AGI
- Examples of expert systems include diagnostic systems, predictive systems, prescriptive systems, decision support systems, and automated reasoning systems.



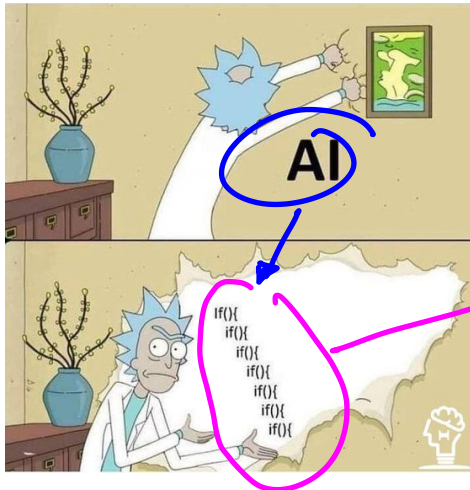
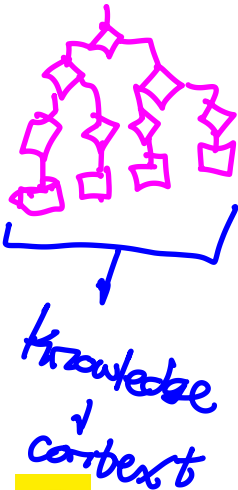
Expert Systems

- An **Expert System** is a system that uses **knowledge** and **reasoning** to solve problems.
- **Expert systems** are used to **automate** and **optimize** decision-making processes.
- Examples of **expert systems** include **diagnostic systems**, **predictive systems**, **prescriptive systems**, **decision support systems**, and **automated reasoning systems**.



Expert Systems as Classical Artificial Intelligence

Here there is a great example of a **diagnostic system**.



decision
tree



Risks and Failures in Information

- **Information systems** are subject to **risks** and failures that can impact business operations. *impact*
- **Risks and failures** can be mitigated through security measures, backup systems, disaster recovery plans, and monitoring tools. *mitigate*
- Examples of risks and failures include security breaches, data loss, system downtime, performance issues, and compliance violations.

incidents



1



2



3



4

observability
real-time
alerts



Risks and Failures in Information

- **Information systems** are subject to **risks** and **failures** that can impact business operations.
- **Risks and failures** can be mitigated through **security measures**, **backup systems**, **disaster recovery plans**, and **monitoring tools**.
- Examples of **risks and failures** include security breaches, data loss, system downtime, performance issues, and compliance violations.

not
scale

all
Capacity
Crash



Outline

1 Information Systems

2 Enterprises as Systems

3 Software Methodologies

[Sub-system]
- information system
expert system



Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to understand the entire organization.

- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.

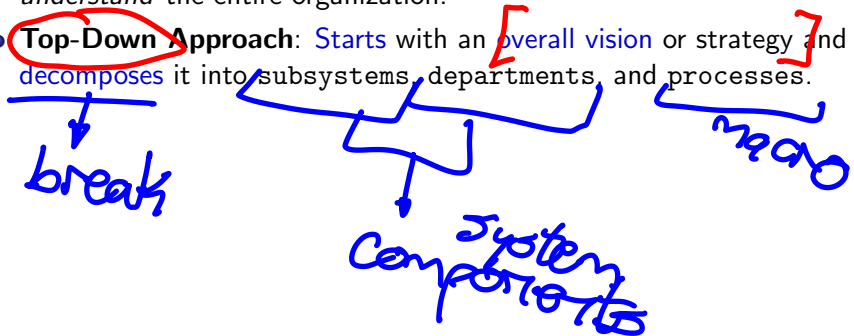
persons
tasks

BPMN
Context



Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to understand the entire organization.
- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.



PIECE Framework for Enterprises

- **Participation** (*Engaging stakeholders*) at every level.
- **Independence of Thought**: Encouraging *diverse, innovative ideas*.
- **Elaboration**: Developing and *refining ideas and processes*.
- **Communication**: Ensuring *clear, effective exchange* of information.
- **Exploration**: Embracing *continuous innovation and improvement*.

Steve Jobs

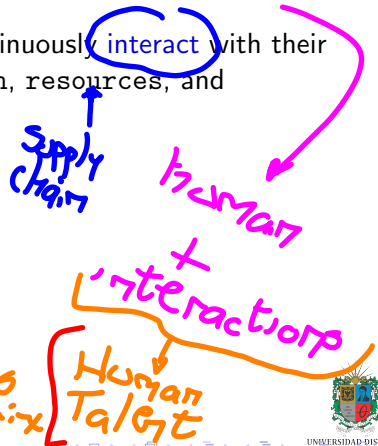
Concepts

strategy
SCRM
BPMN



Enterprise System Typologies

- **Rational Systems:** Organizations driven by logical structured processes and clear hierarchies.
- **Natural Systems:** Organizations viewed as self-organizing entities with emergent behavior.
- **Open Systems:** Enterprises that continuously interact with their external environment for information, resources, and innovation.



Business Systems and Models

- **Business Systems:** Frameworks that encompass an enterprise's internal processes, operations, and strategies. → Data Analysis
- **Examples:** ERP systems, CRM systems, SCM systems.

• **Business Models:** Describe how an organization creates, captures value.

• Examples include subscription-based, freemium, pay-per-use, pay-per-form-based, and direct sales models.

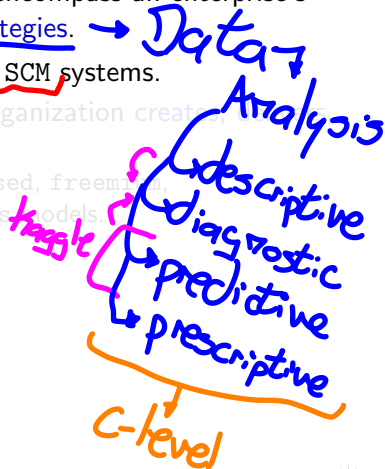
ERP

Enterprise Resource
Planning

CRM

Customer Relationship
Manager

SCM Supply Chain Management



Business Systems and Models

- **Business Systems:** Frameworks that encompass an enterprise's internal processes, operations, and strategies.
- **Examples:** ERP systems, CRM systems, SCM systems.
- **Business Models:** Describe how an organization creates, delivers, and captures value.
 - Examples include subscription-based, freemium, platform-based, and direct sales models.

Netflix

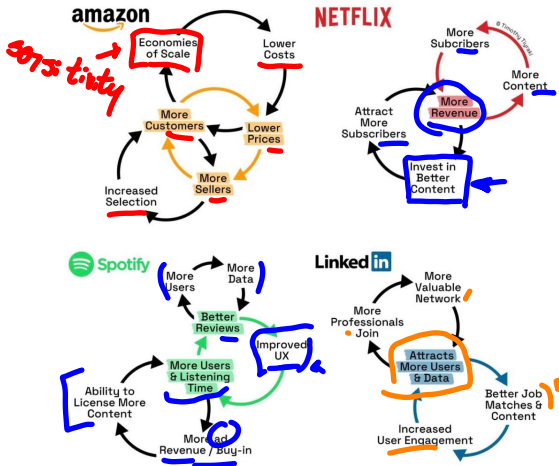
Spotify

Factory



Business Models Examples

Understanding Business Models Through Flywheels



Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Basic Concepts in Software Methodologies

- **Software methodologies** provide frameworks for planning, designing, developing, testing, and maintaining software projects.
- They help teams manage project complexity and ensure quality deliverables.

feasible



Traditional Methodologies ~ 1960-1990

- **Waterfall:** A linear approach where each phase must be completed before moving to the next.
- Suitable for projects with well-defined requirements and low uncertainty.
- Emphasizes thorough documentation and planning.

Requirements

Analysis & Design

Implementation

Testing

Deploy & Maintaining



Traditional Methodologies

- **Waterfall:** A **linear approach** where each phase must be *completed before moving* to the next.
- Suitable for projects with **[well-defined requirements]** and low uncertainty.
- Emphasize thorough **documentation** and **planning**.

20-30 documents

→ rigid



Agile Methodologies

- Emphasize **iterative development**, **customer collaboration**, and **flexibility**.
- Based on the **Agile Manifesto**, which values **individuals** and **interactions** over processes and tools.
- *Examples* include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
- **Agile methodologies** are suitable for projects with **rapidly changing requirements** and **high uncertainty**.
- Promote **adaptive planning**, **evolutionary development**, and **early delivery** of valuable software.



Agile Methodologies

- Emphasize **iterative development**, **customer collaboration**, and **flexibility**.
- Based on the **Agile Manifesto**, which values **individuals** and **interactions** over processes and tools.
- *Examples* include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
- **Agile methodologies** are suitable for projects with **rapidly changing requirements** and **high uncertainty**.
- Promote **adaptive planning**, **evolutionary development**, and **early delivery** of valuable software.



Agile Methodologies

- Emphasize **iterative development**, **customer collaboration**, and **flexibility**.
- Based on the **Agile Manifesto**, which values **individuals** and **interactions** over processes and tools.
- *Examples* include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
- **Agile methodologies** are suitable for projects with **rapidly changing requirements** and **high uncertainty**.
- Promote **adaptive planning**, **evolutionary development**, and **early delivery** of valuable software.



Case Study: Scrum

- **Scrum** employs short, time-boxed iterations called **sprints**.
- Key practices include **daily stand-ups**, **sprint planning**, **reviews**, and **retrospectives**.
- Focuses on **adaptability** and **continuous improvement**.



Case Study: Kanban

- **Kanban** visualizes work items on **boards** and limits Work In Progress (*WIP*).
- Emphasizes **gradual improvements**, **flow management**, and **continuous delivery**.
- Ideal for **projects** requiring **flexibility** with *minimal iteration planning*.



Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

